

Understanding Analysis, 2nd Edition

Chapter 5 - Section 5.2 - Derivatives and the Intermediate Value Property

1. Supply proofs for part (i) and (ii) of Theorem 5.2.4.

Solution: Part (i): $(f + g)'(c) = f'(c) + g'(c)$

Proof:

$$\begin{aligned}(f + g)'(c) &= \lim_{x \rightarrow c} \frac{f(x) + g(x) - f(c) - g(c)}{x - c} = \lim_{x \rightarrow c} \left(\frac{f(x) - f(c)}{x - c} + \frac{g(x) - g(c)}{x - c} \right) \\ &= f'(c) + g'(c)\end{aligned}$$

where the last step is justified because we know both limits exist, so we can apply the Algebraic Limit Theorem (Theorem 4.2.4).

Part (ii): $(kf)'(c) = kf'(c)$ for all $k \in \mathbf{R}$.

Proof:

$$(kf)'(c) = \lim_{x \rightarrow c} k \left(\frac{f(x) - f(c)}{x - c} \right) = k \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = kf'(c)$$

2. Exactly one of the following requests is impossible. Decide which it is, and provide examples for the other three. In each case, assume the functions are defined on all of $\mathbf{R}$.

- (a) Functions f and g not differentiable at zero but where fg is differentiable at zero.
- (b) A function f not differentiable at zero and a function g differentiable at zero where fg is differentiable at zero.
- (c) A function f not differentiable at zero and a function g differentiable at zero where $f + g$ is differentiable at zero.
- (d) A function f differentiable at zero but not differentiable at any other point.

Solution:

- (a) $f(x) = 0$ for $x \geq 0$ and $f(x) = 1$ otherwise. $g(x) = 1$ for $x \geq 0$ and $g(x) = 0$ otherwise. Both functions are discontinuous at zero hence not differentiable there, but fg is the zero function.

- (b) $f(x) = 0$ for $x \geq 0$ and $f(x) = 1$ otherwise. $g(x) = 0$ everywhere.
- (c) Impossible. If $f + g$ were differentiable at zero, then $(f + g) - g = f$ must be differentiable at zero as well by Theorem 5.2.4.
- (d) $f(x) = x^2$ for $x \in \mathbf{Q}$ and $f(x) = 0$ otherwise.

It's differentiable at zero because for $c = 0$, $\frac{f(x)-f(c)}{x-c} = x$ and $\lim_{x \rightarrow 0} x = 0$.

It's not differentiable at $c \neq 0$. Consider a sequence $(x_n) \rightarrow c$. We can split it into two subsequences $(a_n) \in \mathbf{Q}$ and $(b_n) \notin \mathbf{Q}$. We have $(a_n) \rightarrow c^2$ and $(b_n) \rightarrow 0$. Since $c^2 \neq 0$ for $c \neq 0$, the two subsequences converge to different limits, hence (x_n) doesn't converge.

3. (a) Use Definition 5.2.1 to produce the proper formula for the derivation of $f(x) = 1/x$.
- (b) Combine the result in part (a) with the Chain Rule to supply a proof for part (iv) of Theorem 5.2.4.
- (c) Supply a direct proof of Theorem 5.2.4 (iv) by algebraically manipulating the difference quotient for (f/g) in a style similar to the proof of Theorem 5.2.4 (iii).

Solution:

(a)

$$\frac{f(x) - f(c)}{x - c} = \frac{\frac{1}{x} - \frac{1}{c}}{x - c} = \frac{-(x - c)}{xc} \cdot \frac{1}{x - c} = -\frac{1}{xc}$$

So we have $f'(c) = \lim_{x \rightarrow c} (-\frac{1}{xc}) = -\frac{1}{c^2}$ for $c \neq 0$.

(b) We first derive a formula for the derivative of $h(x) = 1/g(x)$.

$$\frac{h(x) - h(c)}{x - c} = \frac{-(g(x) - g(c))}{g(x)g(c)} \cdot \frac{1}{x - c} = -\frac{1}{g(x)g(c)} \cdot \frac{g(x) - g(c)}{x - c}$$

Taking the limit to c and noting that the second term of the product is the definition of $g'(c)$

$$h'(c) = \lim_{x \rightarrow c} -\frac{1}{g(x)g(c)} \cdot \frac{g(x) - g(c)}{x - c} = -\frac{g'(c)}{g(c)^2}$$

Now apply the product rule to $(fh)'(c)$ to get the formula for $(f/g)'(c)$.

— Simpler solution by chatgpt —

Derive the formula of $h'(x)$ by defining $h(x) = 1/x$ then apply the chain rule to $h(g(x))$.

(c)

$$\begin{aligned}\frac{\frac{f}{g}(x) - \frac{f}{g}(c)}{x - c} &= \frac{\frac{f}{g}(x) - \frac{f(x)}{g(c)} + \frac{f(x)}{g(c)} - \frac{f}{g}(c)}{x - c} \\&= \frac{f(x)}{x - c} \left(\frac{g(c) - g(x)}{g(x)g(c)} \right) + \frac{1}{g(c)(x - c)} (f(x) - f(c)) \\&= -\frac{f(x)}{g(x)g(c)} \left(\frac{g(x) - g(c)}{x - c} \right) + \frac{1}{g(c)} \left(\frac{f(x) - f(c)}{x - c} \right)\end{aligned}$$

Take the limit to c to get

$$-\frac{f(c)g'(c)}{g(c)^2} + \frac{f'(c)}{g(c)} = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$$

4. Follow these steps to provide a slightly modified proof of the Chain Rule.

- (a) Show that a function $h : A \rightarrow \mathbf{R}$ is differentiable at $a \in A$ if and only if there exists a function $l : A \rightarrow \mathbf{R}$ which is continuous at a and satisfies

$$h(x) - h(a) = l(x)(x - a) \quad \text{for all } x \in A.$$

- (b) Use this criterion for differentiability (in both directions) to prove Theorem 5.2.5.

Solution:

- (a) First we prove that if such a function l exists then h is differentiable at a . We have

$$h'(a) = \lim_{x \rightarrow a} \frac{h(x) - h(a)}{x - a} = \lim_{x \rightarrow a} l(x) = l(a)$$

Next, we prove that if h is differentiable at a then such a function l exists. Define

$$l(x) = \begin{cases} \frac{h(x) - h(a)}{x - a} & x \neq a \\ h'(a) & (x = a) \end{cases}$$

l is continuous at a since $\lim_{x \rightarrow a} l(x) = \lim_{x \rightarrow a} \frac{h(x) - h(a)}{x - a} = h'(a) = l(a)$.

$h(x) - h(a) = l(x)(x - a)$ for all $x \in A$ by definition.

- (b) TODO.

5. Let $l(x) = \begin{cases} x^a & x > 0 \\ 0 & x \leq 0 \end{cases}$.

- (a) For which values of a is f continuous at zero?
- (b) For which values of a is f differentiable at zero? In this case, is the derivative function continuous?
- (c) For which values of a is f twice-differentiable?

Solution:

- (a) $a > 0$. The limit at 0 from the left is 0. If $a < 0$ then the right limit diverges. If $a = 0$ then the right limit is 1. But if $a > 0$ then the right limit is 0.

- (b) From the direction of positive x , we have $\lim_{x \rightarrow 0^+} \frac{x^a - 0^a}{x - 0} = \lim_{x \rightarrow 0^+} x^{a-1} = 0^{a-1}$. This is only defined for $a > 1$, in which case it's 0 and agrees with the left limit, so the derivative is defined.

The derivative is continuous, but seems to require us to use the fact that $l'(x) = ax^{a-1}$ for *non-integer* a , which we have not proved yet.

- (c) The next right-side derivative at zero is $\lim_{x \rightarrow 0^+} x^{a-2} = 0^{a-2}$, so we need $a > 2$.